

Spatial aggregation confounds null model-based inference of species association networks

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1 Abstract

2 Introduction

3 Species interactions are fundamental to the evolutionary origins of biodiversity and the main-
4 tenance of that diversity in ecological assemblages (Hutchinson 1961, Holt and Lawton 1994,
5 Chesson 2000, Proulx et al. 2005, Bascompte and Jordano 2007, Levine and HilleRisLambers
6 2009, Allesina and Tang 2012, Johnson and Bronstein 2019). Network theory has come to be
7 a central paradigm in ecology and evolution (May 1972, Williams and Martinez 2000, Dunne
8 et al. 2002, Proulx et al. 2005, Bascompte and Jordano 2007), and so, not surprisingly, these
9 interactions are often cast as ecological networks where species are nodes (or vertices) and
10 network links (or edges) connect species that interact.

11 Abundance or presence/absence data are among the most prevalent and accessible ecologi-
12 cal data, and have consequentially been frequently leveraged (Morueta-Holme et al. 2016,
13 Calatayud et al. 2019) in an attempt to infer species interactions (or more accurately as-
14 sociations), beginning at least in the work of Diamond (1975). The specific data used for
15 constructing association networks are often site by species matrices where each cell in the
16 matrix records the abundance (or presence/absence) of a given species at a given site (i.e. sam-
17 pling location). For simplicity I will refer to both abundance and presence/absence data as
18 simply “abundance data” and focus on true abundance data with the assumed implication
19 that methodological limitations in the case of true abundances will be even more confounding
20 for presence/absence scenarios (Ulrich and Gotelli 2010).

21 Two salient insights from decades of debate about using abundance data to infer ecological
22 processes are: 1) interactions in the strict sense cannot be inferred, but *species associations*
23 might be recoverable and these *associations* are assumed to map onto a combination of species-
24 species and species-environment interactions (Morueta-Holme et al. 2016, Calatayud et al.

25 2019); and 2) robust null models are necessary to identify which patterns in the data are likely
26 to arise by chance and which are likely to carry biological signal (Ulrich and Gotelli 2010).

27 Null models seek to generate plausible data, and patterns in data, while stripping away the
28 actual relationships between species in the data (Ulrich and Gotelli 2010). Most often a null
29 model is achieved by repeatedly permuting the original data such that any real associations
30 are removed because which species co-occur has been randomized by permutation (Ulrich
31 and Gotelli 2010). The collection of all such permutations represents a null distribution.
32 These permutation approaches also seek to preserve key aspects of the original data such
33 as maintaining the observed abundances of each species and/or maintaining the observed total
34 number of individuals at each site (Ladau 2017). These constraints are imposed in an attempt
35 to control the rate of incorrect inference of the presence or absence of network links (Ladau
36 2008, 2017). Other methods for inferring association networks do exist, especially model-based
37 inference applied in microbial ecology (e.g. Zhou et al. 2010, Kurtz et al. 2015), but are less
38 frequently employed in community ecology (but see, e.g., Harris 2016, Losapio et al. 2021).

39 Here I raise a previously unreported technical issue with the approach of inferring association
40 networks from abundance data using permutational null models. The issue arises from the
41 near universal spatial aggregation of individuals within species (Legendre 1993, Condit et al.
42 2000, Lennon 2000, Dale and Fortin 2002, McGill and Collins 2003, Engen et al. 2008, Zillio
43 and He 2010, Harte 2011, Connolly et al. 2017). Using simulation experiments and analysis of
44 real data (from Calatayud et al. 2019), I show that intra-specific spatial aggregation (ISSA)
45 causes exceedingly high type I error rates in permutational null model approaches to inferring
46 association networks, i.e., incorrectly detecting associations when there are none. ISSA in itself
47 is not evidence of species association—it can arise from a large number of processes including
48 simple demographic drift (Kendall 1949, Hubbell 2001) in which species do not meaningfully
49 interact with each other or shared environments.

50 In the absence of ISSA, abundances of a species across sites should follow a Poisson distribu-
51 tion (Diggle 2003). Thus ISSA is often quantified as the deviation of observed data from a
52 Poisson distribution (Diggle 2003). Conversely, the negative binomial distribution can model
53 spatially aggregated intra-specific abundances across sites (Bliss and Fisher 1953, Taylor 1961,
54 Engen et al. 2008, Connolly et al. 2017). Critically, the negative binomial arises very naturally
55 from ecological processes as the distribution of population sizes in a birth-death-immigration
56 model (Kendall 1949). If the abundance of a species at a site is determined by local birth
57 and death with supplemental individuals arriving by immigration from a source pool, then
58 the probability distribution of repeatedly sampled abundances will follow a negative binomial
59 distribution (Kendall 1949). In this way, simple demographic drift (birth, death, and immi-
60 gration that are purely stochastic) is expected to produce intra-specific abundances that are
61 negative binomial-distributed and spatially aggregated. Thus ISSA is not, by itself, evidence
62 of species association.

63 The technical issue raised here is specific to the widely used family of methods that infer
64 association networks by comparing an observed test statistic to a null distribution generated
65 by Monte Carlo permutation of the site-by-species matrix. I do not test whether the same

issue affects methods that do not rely on this style of null model, most notably joint species distribution models, which explicitly condition on measured or latent environmental covariates and site-level random effects (e.g. Zhou et al. 2010), or the compositional correlation methods common in microbial ecology (e.g. Kurtz et al. 2015). It is not obvious that the mechanism described here transfers directly to these other methods, and I do not want to imply, by omission, that it does. However, any effects of ISSA on joint species distribution modeling and compositional correlation methods should be assessed. Here I focus solely permutational null models because of their foundational status in the field (**cite?**) and their continued use in inferring association networks.

Methods

Reproducibility and data

A custom R package, *SpatAggNet* accompanies this paper and can be used to reproduce all results. `ssec{sec-setup}` details how to access and install *SpatAggNet*. All analysis scripts and data analyzed are available via the package.

I use a compiled data set of site-by-species abundance matrices assembled by Calatayud et al. (2019). The data are included with the *SpatAggNet* package as `abund_mats` and comprise `|r length(abund_mats)|` site-by-species matrices. These compiled data are ideal for investigating the impact of ISSA on association network inference because they were used by Calatayud et al. (2019) for the very purpose of network inference and also because they represent a spatially and taxonomically extensive collection of quality-controlled data.

Quantifying spatial aggregation

The spatial species abundance distribution (SSAD) is the distribution of a single species' replicate abundances across sites. I fit both negative binomial and Poisson distributions to the SSADs of each species in each site-by-species matrix using maximum likelihood. The `nb_fit()` function from *SpatAggNet* fits both distributions to every species in each site-by-species matrix and returns, for each species from each matrix:

- the negative binomial dispersion parameter (`size`, i.e. k) and mean (`mu`)
- the log-likelihoods of negative binomial and Poisson models
- their AIC difference (`delta_AIC`),
- a χ^2 -distributed goodness-of-fit statistic for the negative binomial [`zsq`; Etienne (2007)]
- the species' total abundance and occupancy.

97 **Analysis of empirical data**

- 98 • purpose: show the issue and that nb is the reason
- 99 • data from rareplusminus
- 100 • get sad and ssad from those data
 - 101 – pois
 - 102 – nb
 - 103 – model selection of pois and nb
- 104 • look at network stats for real data versus data simulated from sad and ssad
 - 105 – pois: doesn't look like real data
 - 106 – nb: looks close to real data
- 107 • report on rareplusminus stuff

108 **Simulation experiment**

- 109 • purpose: show the issue persists across different methods
- 110 • start with data simulated from sad and ssad
 - 111 – only need to do nb
- 112 • make networks based on diff perm methods and diff distance metrics

113 **Behavior of negative binomial random variables subjected to permutation**

- 114 • show that k is distorted

115 **Results**

116 **Analysis of empirical data**

117 **Simulation experiment**

118 **Discussion**

119 **References**

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